

Born Robust, Then Calibrated

A Two-Timescale Theory of Continuous-Attractor Formation
Across the Innate–Learned Hand-off

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Abstract

Continuous attractor networks (CANs) underlie neural codes for head direction, grid, and place. Their textbook weakness is the *fine-tuning problem*: an exact continuum of fixed points is a measure-zero, structurally fragile object, so any weight asymmetry makes the bump drift. Two recent results put this framing under pressure. First, robustness theory shows the brain need only sit *near* a slow invariant manifold, not on an exact attractor [14]. Second, developmental data show that ring and toroidal *topology* appears before eye opening and before navigation [2, 9], while accurate, gain-1 path integration requires an experience-dependent learning phase [18]. We propose a theory that reconciles these: **the topology of the attractor is an innate, structurally stable property of a normally hyperbolic slow manifold, whereas the metric along the manifold — the path-integration gain and the anchoring to the world — is the only fine-tuning-sensitive quantity, and is set by local plasticity.** The fine-tuning problem thus splits into a robust part (topology, genetically scaffolded) and a fragile part (gain/drift, learned). We will (0) show that the scaffold itself can be built before experience, by spontaneous activity acting on a generic distance-biased connection rule, so that only its connectivity *statistics* — not its individual weights — need be innate; (i) show that a coarse innate scaffold yields a structurally stable ring/torus slow manifold that deforms smoothly while acquiring a leading-order on-manifold drift; (ii) derive a three-factor, self-supervised plasticity rule that calibrates gain to 1, anchors the manifold to landmarks, and *continuously* repairs drift — unifying the “build” and “maintain” literatures; and (iii) turn the theory into falsifiable predictions about the P10–P21 developmental trajectory and about adult plasticity blockade. The program targets three conceptual gaps the field has left open: the innate/learned hand-off, whether an exact continuous attractor is even the right target, and the separation of learning rules from robustness mechanisms.

1 Background and the central tension

CANs explain the head-direction ring, the grid/place torus, the oculomotor line integrator, and bump working memory [10, 3]. The classical construction assumes translation-invariant connectivity $W(\mathbf{x}, \mathbf{y}) = f(\mathbf{x} - \mathbf{y})$; a continuum of marginally stable states then exists only on a measure-zero surface in parameter space, so small asymmetries destroy it [14, 12]. This is the fine-tuning problem.

The field has produced two largely separate responses. A *learning* thread grows the required connectivity by local plasticity — trace/Hebbian rules [17], asymmetric STDP for grids [19], and a local self-supervised rule that learns the *Drosophila* ring connectome and attains gain-1 integration [18]. A *robustness* thread shows that imperfect attractors behave well enough: homeostatic plasticity restores the tuning condition [8], short-term plasticity pins the bump and suppresses drift [15], and continuous-symmetry dynamics survive disordered embeddings [4]. Crucially, Park, Ságodi et

al. [14] reframe the whole problem: what matters behaviorally is proximity to a *slow invariant manifold*, not exactness.

These threads collide with development. Head-direction attractor coupling is present at $\sim$ P12, before stable tuning and before eye opening [2]; toroidal grid topology is reported as early as P10, before navigation [9]; grids still form under geometric deprivation, though stabilization needs experience (PNAS 2023, via [9]). Yet accurate path integration — gain exactly 1 — demands a developmental self-supervised phase [18]. **No theory specifies the hand-off**: what is genetically scaffolded, what plasticity refines, and how a “preconfigured” attractor avoids the fine-tuning problem without experience (Review §6).

2 Hypothesis: topology is robust, the metric is learned

Our organizing claim separates two things the literature conflates:

H1 (Topology is innate and robust). A coarse, genetically specifiable center-surround scaffold produces a *normally hyperbolic* slow manifold $\mathcal{M}$ with ring (1D) or toroidal (2D) topology. By invariant-manifold theory [6], $\mathcal{M}$ persists under C^1 -small perturbations of the connectivity: its topology is structurally stable, *not* fine-tuned.

H1' (Statistics, not structure — relaxed form). The scaffold need not be a clean, translation-invariant center-surround kernel. It suffices that W_0 possess the right *second-order statistics*: continuous-attractor dynamics with a continuous symmetry arise from random, disordered embeddings [4] and from low-rank connectivity $W_0 \approx \sum_k \mathbf{a}_k \mathbf{b}_k^\top$ with appropriately correlated random vectors [13]. The required “structure” is then a property of connectivity *statistics*, which a generic developmental rule can set, rather than a hand-built kernel.

H1 is the strong, didactic form; H1' is what the theory actually needs and what Aim 0 supplies.

H2 (The metric is fragile and must be learned). Sufficiently small perturbations deform $\mathcal{M}$ into a nearby manifold and induce a slow *flow along* it. The computation-sensitive quantities are (a) the path-integration gain g (whether self-motion advances the bump at the right rate) and (b) the anchoring of the intrinsic coordinate to the world. These are what three-factor, self-supervised plasticity sets.

Under H1–H2 the fine-tuning problem dissolves by decomposition. The fragile, measure-zero condition is not “does an attractor exist” but “is the on-manifold flow null and correctly geared.” Topology can be born; gain and anchoring must be calibrated. This predicts precisely the observed developmental order — manifold first, calibrated behavior later — and unifies the build and maintain threads, because the same rule that calibrates gain also continuously cancels drift.

3 Model

We study a rate network of N neurons with preferred phases $\theta_i \in S^1$ (ring; the torus is the product of two such factors):

$$\tau \dot{\mathbf{r}} = -\mathbf{r} + \phi(\underbrace{W_0}_{\text{innate}} + \underbrace{\Delta W(t)}_{\text{plastic}}), \quad W = \underbrace{W_0}_{\text{innate}} + \underbrace{\Delta W(t)}_{\text{plastic}}. \quad (1)$$

ϕ is a bounded, monotone nonlinearity. The innate scaffold splits into a symmetric and an anti-symmetric part,

$$W_0 = W_0^s + g_0 W_0^a, \quad W_0^s(\theta_i, \theta_j) = K(\theta_i - \theta_j) \text{ (even)}, \quad W_0^a = K'(\theta_i - \theta_j) \text{ (odd)}, \quad (2)$$

where K is a coarse center-surround kernel. The even part W_0^s supports a bump and hence a manifold $\mathcal{M} = \{\mathbf{r}^*(\psi) : \psi \in S^1\}$ of bump states; the odd part couples to angular velocity $\omega(t)$ (carried in $\mathbf{I}$) to translate the bump. The scaffold is *coarse*: it fixes topology but *not* the gain g_0 , which is generically $\neq 1$.

What needs calibrating. Driving (1) with angular velocity $\omega(t)$ moves the bump coordinate as

$$\dot{\psi} = g \omega(t) + v_{\text{drift}} + \sqrt{2D} \xi(t), \quad (3)$$

with gain g , deterministic drift v_{drift} , and diffusion D . Accurate integration requires $g = 1$ and $v_{\text{drift}} \approx 0$. The plastic term ΔW has two roles: a path-integration component ΔW^a that sets g and cancels v_{drift} , and an anchoring component ΔW^L that ties ψ to landmark input.

Three-factor self-supervised plasticity. Following the logic of [18], the network is its own teacher: it compares the bump displacement it *produces* against the displacement *implied* by a copy of self-motion, and corrects the asymmetric weights to null the mismatch. Schematically, with prediction error $e(t) = \hat{\psi}_{\text{sensed}} - \dot{\psi}_{\text{network}}$,

$$\Delta \dot{W}_{ij}^a = \eta e(t) \partial_{\theta} r_i r_j - \gamma \Delta W_{ij}^a, \quad \Delta \dot{W}_{ij}^L = \eta_L (r_i - \langle r \rangle) \ell_j, \quad (4)$$

where ℓ_j is landmark drive and γ is homeostatic decay. The $\partial_{\theta} r_i r_j$ factor is an odd activity correlation that projects learning onto the translation (Goldstone) mode. The implementation assumes a population-level error and computes $\partial_{\theta} r_i$ from ordered neighbouring units; a biological circuit supplying those factors remains to be specified. No backpropagation or target-angle tutor is used.

4 Aim 0 — Where the innate scaffold comes from

Goal. Remove the appearance that a structured W_0 is assumed from nowhere, by deriving it — up to its statistics — from mechanisms that act *before* structured experience.

Two routes, one requirement. H1' says only the second-order statistics of W_0 matter. We will show two biologically grounded mechanisms that supply those statistics without specifying individual synapses.

(0a) *Genetic connection rule.* A distance-dependent connection probability $p_{\text{con}}(\theta_i, \theta_j) \propto \rho(|\theta_i - \theta_j|)$ — the generic outcome of axonal/dendritic arbor geometry and molecular gradients — yields, in expectation, an even center-surround kernel plus quenched disorder. Individual weights are random; the *statistics* are genetic. The *Drosophila* ring attractor, whose connectivity is connectome-specified [11], is an existence proof.

(0b) *Self-organization from spontaneous activity.* Before eye opening, internally generated correlated activity drives an unsupervised, symmetry-seeking plasticity (Hebbian on a locally biased substrate) that sculpts W_0^s . We will model a pre-experiential “Phase 0” in which the same machinery that later calibrates (Aims 1–2) first *builds* the manifold, as in the STDP construction of [19].

Claim 0.1. A generic distance-biased connection rule (0a), or Phase-0 self-organization (0b), produces connectivity whose statistics place the network in the bump-forming, normally hyperbolic regime of Aim 1 with probability $\rightarrow 1$ as $N \rightarrow \infty$ — i.e. the robust part of the attractor is *cheap* and requires no fine-tuning of individual weights.

Deliverable / prediction. Simulated Phase-0 yields ring/torus topology *before* any structured input, with the gain still uncalibrated ($g \neq 1$). This is the model correlate of attractor coupling and toroidal topology appearing before eye opening and navigation [2, 9], and it predicts that disrupting *spontaneous activity* (not sensory experience) is what abolishes the manifold.

5 Aim 1 — Topology is structurally stable; only the metric is fragile

Goal. Make H1–H2 precise and prove the decomposition.

Claim 1.1 (persistence). For the innate scaffold W_0 with a spectral gap between the slow (bump-translation) mode and the contracting bulk of the *dynamics’ Jacobian* $J(\psi) = -I + \text{diag}(\phi')W_0$, the slow manifold $\mathcal{M}$ is normally hyperbolic, and by [6] a nearby invariant manifold $\mathcal{M}_\epsilon$ of the same topology persists for every C^1 -small ΔW . *Topology does not require fine-tuning.*

Claim 1.2 (drift is the computationally sensitive quantity). A generic perturbation ΔW deforms the persistent manifold transversally by $O(\|\Delta W\|)$ and also induces an on-manifold flow. Normal hyperbolicity controls the former; the leading rate of the latter is the projection of ΔW onto the left zero-eigenvector (Goldstone mode) $\mathbf{n}(\psi)$ of $J(\psi)$:

$$v_{\text{drift}}(\psi) = \mathbf{n}(\psi)^\top \Delta W \mathbf{r}^*(\psi) + \mathcal{O}(\|\Delta W\|^2). \quad (5)$$

Hence the measure-zero, fine-tuning-sensitive condition is $v_{\text{drift}} \equiv 0$ and $g = 1$ — a property of the *flow on $\mathcal{M}$* , not of its existence.

Why this avoids the standard pitfalls. The sensitive object is the Goldstone mode of the *Jacobian* J , not an eigenvalue of W ; existence of $\mathcal{M}$ is bought by normal hyperbolicity (a robust, open condition), not by symmetry of W . We therefore do not smuggle the translation symmetry back in through the inputs.

Deliverable. A perturbation bound: topology survives for $\|\Delta W\| < \delta_{\text{top}}$ (open set, $O(1)$ size), while drift grows linearly in $\|\Delta W\|$ (5). This quantifies exactly how much “innate sloppiness” is tolerable and how much residual error plasticity must remove.

6 Aim 2 — One local rule that calibrates, anchors, and maintains

Goal. Show the rule (4) drives $g \rightarrow 1$, $v_{\text{drift}} \rightarrow 0$, and anchors ψ , and that it keeps doing so against ongoing synaptic noise — i.e. the same plasticity *builds* and *maintains*.

Analysis. Under timescale separation ($\eta \ll 1/\tau$), average (4) over the velocity process to obtain a deterministic flow for the slow variables (g, v_{drift}). The self-supervised error $e(t)$ makes $g = 1$, $v_{\text{drift}} = 0$ the *unique* stable fixed point of the averaged gain dynamics, with the homeostatic term γ bounding the weights. We will prove convergence *with explicit, checkable conditions* — ergodic, zero-mean-free velocity statistics; persistent excitation of ω ; $\eta \ll 1/\tau$; bounded ϕ — rather than invoking a generic stochastic-approximation theorem and asserting convergence. We will also characterize the failure modes (degenerate velocity statistics $\Rightarrow$ unidentifiable gain; too-large $\eta \Rightarrow$ oscillatory weights).

Continuous repair (unifying §2 and §4 of the review). Add synaptic fluctuation $\Delta\dot{W} += \sqrt{2\sigma_W^2}\zeta(t)$ as in [14]. Because the calibration fixed point is *attracting*, the same rule that built the gear continuously cancels the drift that fluctuations re-inject; we will compute the steady-state residual drift/diffusion as a function of σ_W and η and compare to the passive short-term-plasticity pinning of [15]. This yields the missing unified “built *and* maintained” account.

7 Aim 3 — Developmental predictions and falsification

The theory makes the developmental order a *consequence*, not an assumption, and produces discriminating experiments.

1. **Order of emergence.** Topology (manifold, attractor coupling) $\rightarrow$ calibrated gain (accurate path integration) $\rightarrow$ landmark anchoring. *Matches* HD coupling before stable tuning [2] and toroidal topology before navigation [9]; *predicts* that intrinsic manifold structure is measurable (persistent homology of population activity) before behavioral gain reaches 1.
2. **Deprivation dissociation.** Manipulating early experience should perturb *gain accuracy and anchoring* but leave *manifold topology* intact — consistent with grids forming under geometric deprivation while stabilization is impaired. Topology is the structurally stable part (Aim 1); only the learned metric should be experience-sensitive.
3. **Drift decreases over development.** The model predicts a measurable decline of v_{drift} and a rise of g toward 1 across P10–P21, with a quantitative rate set by velocity-experience statistics (Aim 2). This is a direct, time-resolved fit target — the methodological gap the review flags (no model is fit to the P10–P21 trajectory).
4. **Adult plasticity blockade — the key discriminator.** If the same rule continuously re-tunes (our claim), then *blocking the plasticity in the adult* should cause slow gain drift and a gradual return of the fine-tuning problem. If instead the attractor is “built-then-frozen,” adult blockade is inert. This single experiment distinguishes “continuously re-tuned” from “built-then-maintained.”

8 Methods

Spontaneous-activity phase (Aim 0). Initialise connectivity from a distance-biased random connection rule; run a Phase-0 of internally generated correlated activity under Hebbian/STDP plasticity; verify emergence of ring/torus topology (persistent homology) before any structured input. **Simulation.** Integrate (1)–(4) for $N = 256$ (ring) and $N = 64^2$ (torus); innate K from a coarse center-surround; self-motion inputs with naturalistic angular-velocity statistics. **Manifold/topology.** Estimate intrinsic dimension and topology by persistent homology and PCA/Isomap on population activity (as used to confirm the toroidal code [7]). **Stability geometry.** Compute the Jacobian $J(\psi)$ along $\mathcal{M}$, its Goldstone mode, and Lyapunov exponents transverse vs. tangent to $\mathcal{M}$ to verify normal hyperbolicity (Aim 1). **Drift/gain readout.** Fit (3) by regressing bump displacement on ω ; track $g(t)$, $v_{\text{drift}}(t)$, $D(t)$ over learning. **Data contact.** Compare emergence order and drift-vs-age curves to [2, 9]; the design-theoretic constructions of MADE [5] provide an analytic check on which innate scaffolds give which topology.

9 What is new, and relation to prior work

- **Vs. learning models** [17, 19, 18]: they grow the *whole* attractor from scratch. We instead *assume the topology is innate* (justified by [9, 2]) and assign plasticity the narrower, well-posed job of metric calibration — which is also why our rule is identifiable and provably convergent.
- **Vs. robustness theory** [14, 15, 8]: they show approximate attractors suffice but treat the manifold as given. We supply the *developmental origin* of that manifold and make robustness and learning the same mechanism.
- **Vs. the spectral/normative route** [16]: rather than deriving a manifold from task optimization, we exploit the structural stability of normally hyperbolic manifolds to explain why topology can be *cheap and innate*, and locate all the genuinely hard tuning in a one-dimensional gain.

The conceptual payoff: the fine-tuning problem was always stated about the wrong object. Existence of the attractor is robust; only its *gearing* is fragile, and gearing is exactly what experience-dependent plasticity is good at calibrating.

10 Risks and alternatives

R1. Normal hyperbolicity may fail for biologically coarse scaffolds (weak spectral gap). *Mitigation:* quantify the minimal gap; test whether inhibitory recurrence à la [3] supplies it. **R2.** The self-supervised error signal $e(t)$ may lack a clear neural substrate. *Mitigation:* adopt the concrete signals identified by [18]; treat the substrate as a prediction to be tested. **R3.** Topology might *not* be fully innate. *This is a feature:* Aim 3’s deprivation and blockade experiments are designed to falsify H1 if the manifold turns out to be experience-built.

11 Timeline

Months 1–4: Aim 0 — spontaneous-activity / connection-rule origin of the scaffold; establish H1’ numerically. **Months 3–9:** Aim 1 theory + simulation of the innate scaffold. **Months 7–15:** Aim 2 plasticity analysis and convergence proof. **Months 13–20:** Aim 3 developmental fits and predictions; manuscript.

A Design rationale: is a structured W_0 a strong assumption?

A standing note recording why the innate scaffold is a weak — not strong — assumption, and where its structure comes from. It collects the reasoning behind H1’ and Aim 0 so the argument can be revisited.

The assumption splits into a robust part and a fragile part. W_0 bundles two logically distinct things. (i) The *motif* — local excitation with broader inhibition (center-surround) — which creates the bump and hence the ring/torus topology. (ii) The *homogeneity* — that the kernel is identical for every neuron (exact translation invariance) — which makes the manifold perfectly flat. Only (i) is assumed, and (i) is robust: it holds on an *open set* of parameters (Amari’s local-excitation/lateral-inhibition bump [1]), which is exactly why Aim 1 rests on normal hyperbolicity, an open and structurally stable condition. Property (ii) is *measure-zero*, and the

proposal deliberately does *not* assume it: the inhomogeneity of W_0 is precisely what appears as the uncalibrated gain $g_0 \neq 1$ and residual drift in (3), and is handed to plasticity. Assuming (ii) would re-import the fine-tuning problem; assuming only (i) does not.

Where the motif comes from. Two non-circular origins, both predicting the observed developmental timing. (a) *Genetic specification of a rule, not of weights*: a distance-dependent connection probability — the generic consequence of arbor geometry and molecular gradients — yields the motif in expectation without specifying N^2 synapses; the *Drosophila* ring attractor is a connectome-specified existence proof [11]. (b) *Self-organization from spontaneous activity*: before eye opening, internally generated correlated activity can sculpt the motif via Hebbian plasticity on a locally biased substrate [19]. Either way the regress bottoms out in cheap, generic mechanisms, and (b) explains why the manifold appears *before* experience [2, 9].

Statistics, not structure. The requirement on W_0 is weaker than “center-surround.” Continuous-symmetry dynamics arise from random, disordered embeddings [4] and from low-rank connectivity with correlated random vectors [13], so what must be innate is a set of connectivity *statistics*, not a hand-built kernel (H1’). This is what makes the genetic and spontaneous origins above plausible: a generic rule can set statistics, not exact weights.

The bootstrapping bound. The relaxation is not free. Plasticity can calibrate the gain only if the self-supervised error (4) is meaningful, which requires the uncalibrated manifold to be *slow enough* that the bump stays coherent on the learning timescale. Too inhomogeneous a W_0 scatters the bump before learning can act. This places W_0 in a “coarse-but-coherent” window — an upper bound on disorder set by the same δ_{top} -versus-drift trade-off as Aim 1. Quantifying that window is itself a target result.

Why this is not circular. We do not derive structured connectivity from nothing. We *relocate the fine-tuning bottleneck*: the structurally stable topology comes from cheap, generic mechanisms (genetics or spontaneous activity), while the only measure-zero, experience-sensitive quantity is a one-dimensional gain. The contribution is to identify *where* the hard tuning actually lives, and to show that the robust part need not be tuned.

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